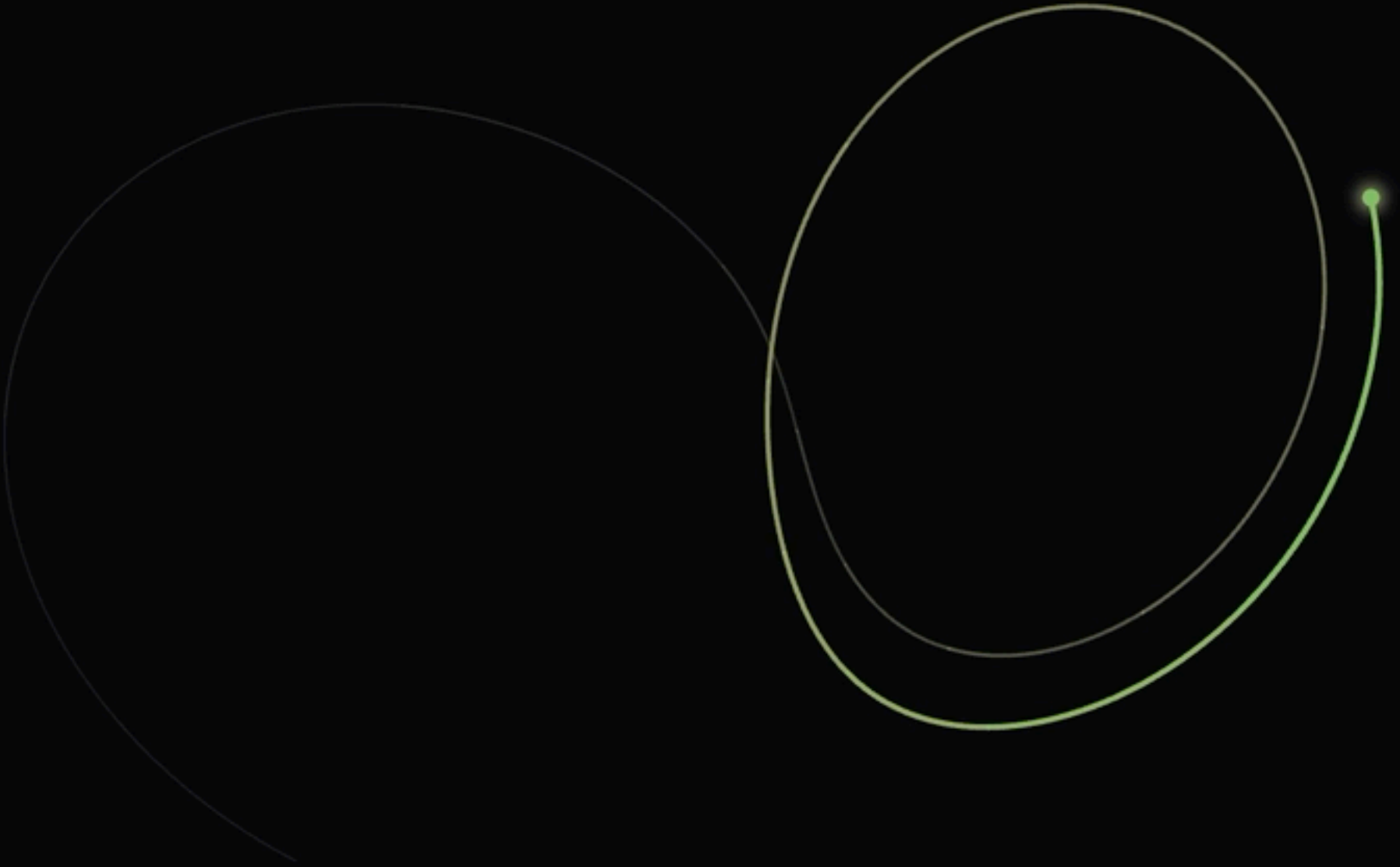


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- $x_0 = 10.00000000$
- $x_0 = 10.00001000$
- $x_0 = 10.00002000$



TIME
1.52

SEPARATION
 1.45×10^{-5}

CHAOS THRESHOLD r_H
24.74
r is above it — chaotic

Pause

Restart

INITIAL GAP

10⁻⁵

SPEED

0.8x

r

28.0

σ

10.0

b

2.67

TRAIL

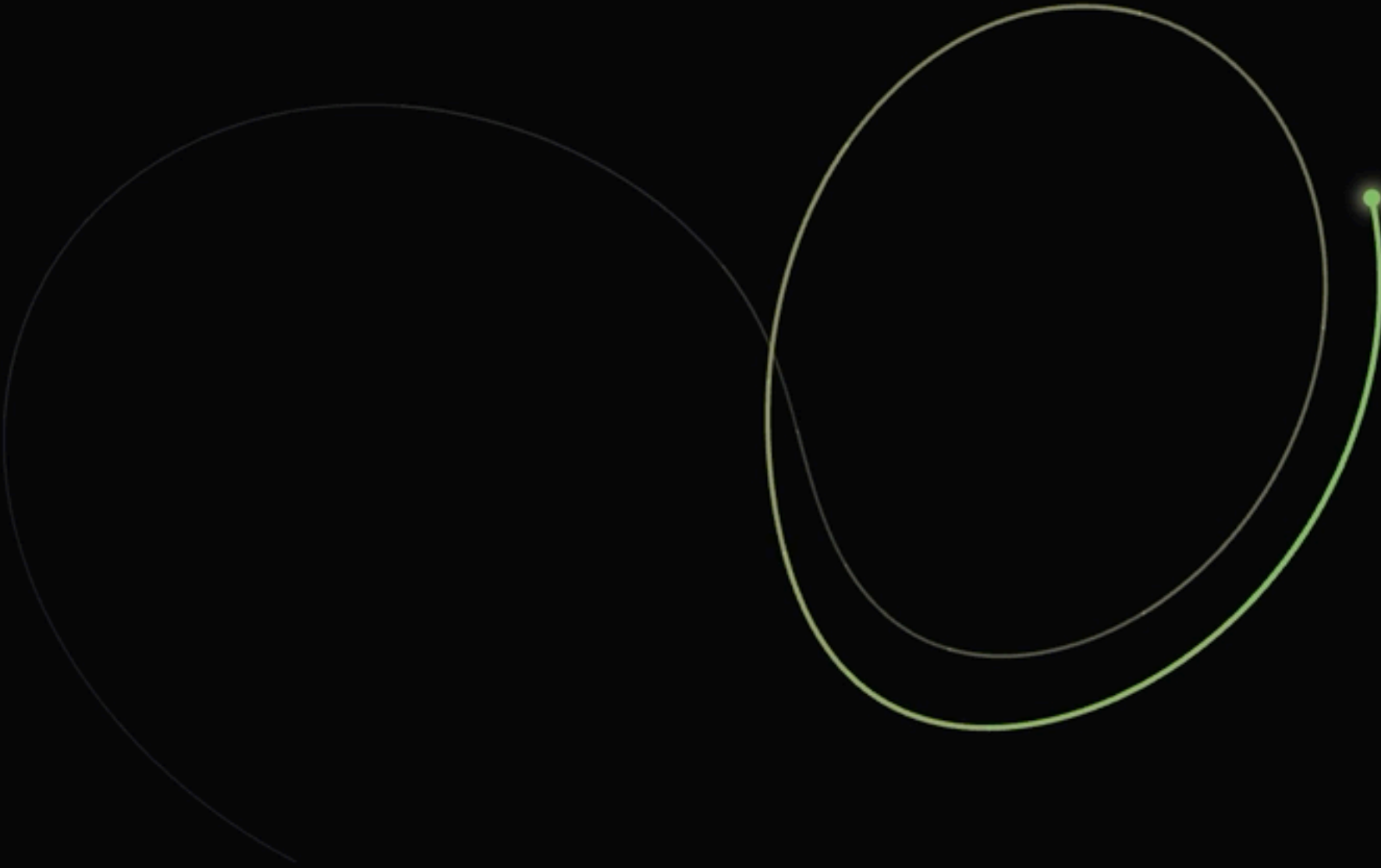
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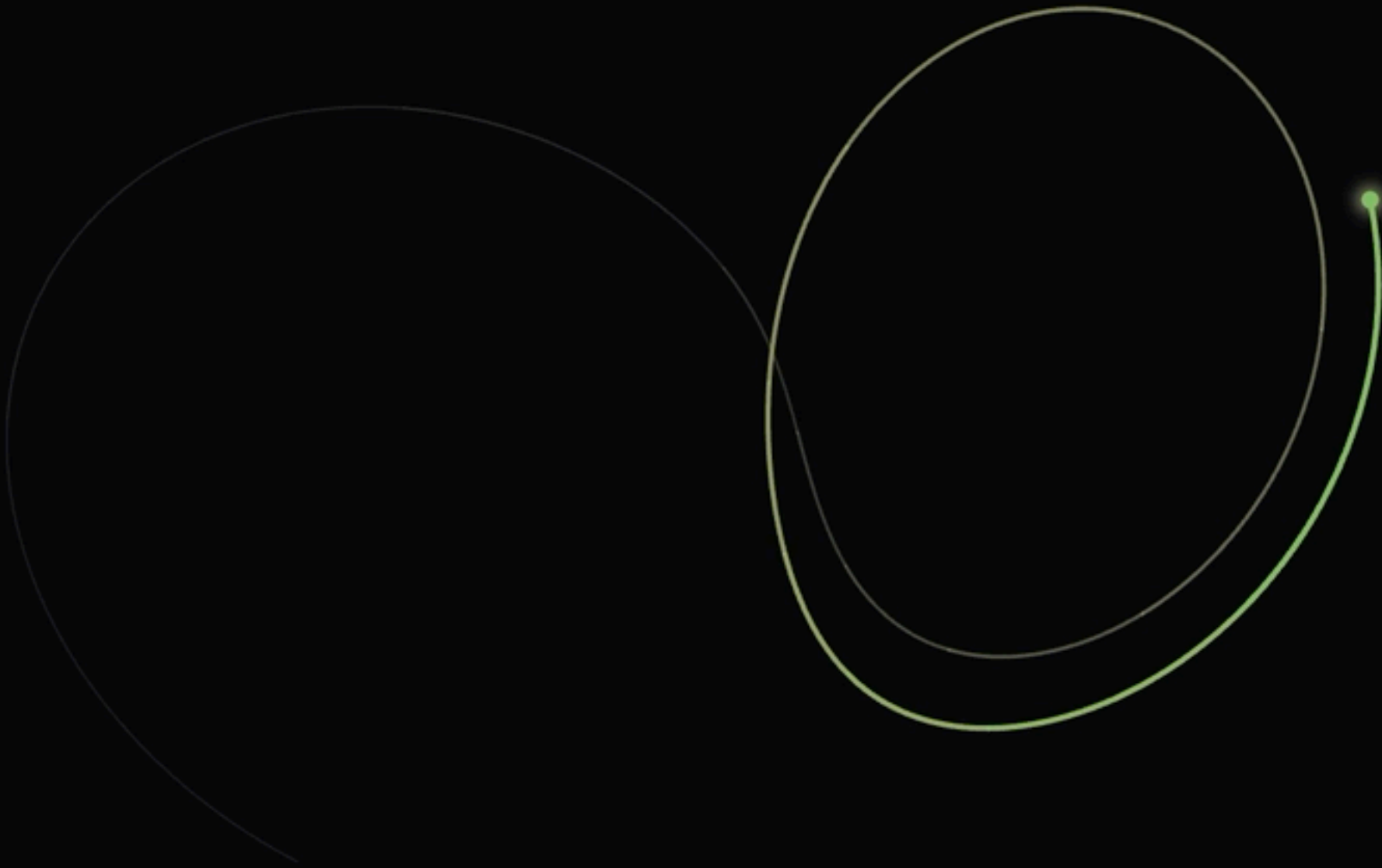
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Trapping Lemma for the Lorenz System

Let

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(r - z) - y, \\ \dot{z} &= xy - bz.\end{aligned}$$

Lemma. *There exists a bounded ellipsoid E that is a trapping region for the Lorenz flow. Equivalently, the time-one map T of the Lorenz flow satisfies*

$$T(E) \subset \text{int}(E).$$